



PSORIASIS

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Morphological
terms



1ry lesions



2ry lesions

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PRIMARY LESIONS

- Macule : change in color (flat , circumscribed) < 1 cm .
- Patch : change in color (flat , circumscribed) > 1 cm .



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- Papule : elevated circumscribed lesion < 1 cm
flat topped , umblicated , verrucous dome shaped
- Plaque: elevated circumscribed lesion > 1 cm



- 5 • nodule : elevated palpable circumscribed >1 cm
with dermal or sc component
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- Wheal : transient elevation of skin due to dermal oedema



- Vesicle : elevated circumscribed lesion filled with fluid <1 cm



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- Bullae : elevated circumscribed lesion filled with fluid $> 1\text{ cm}$
- Pustule : elevated circumscribed lesion filled with purulent fluid $< 1\text{ cm}$



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2RY LESIONS

- Crust :dried serum (blood or pus) on surface of skin .
- Scale : hyperkeratosis accumulation of stratum corneum .
- Fissure : linear cleft in the skin



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- Excoriation : exogenous injury to all or part of epidermis
- Erosion : partial loss of epidermis
- Ulcer : total loss of epidermis , may have loss of dermis



9 PAPULOSQUAMUS DISEASE



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PSORIASIS



II WHAT IS PSORIASIS

- Psoriasis is a chronic, immune-mediated disorder that results from a genetic predisposition combined with environmental triggers characterized by sharply demarcated erythematous scaly plaques.
- • Psoriasis is a systemic disease that may be associated with psoriatic arthritis, metabolic syndrome, and atherosclerotic cardiovascular and other diseases.

12 EPIDEMIOLOGY

- 2% of world population
- Two peaks in age of onset have been reported (see later):
 - Early-onset psoriasis: at 20-30 years of age.
 - Late-onset psoriasis: at 50-60 years of age.
- Equal in both sex

13 AETIOPATHOGENESIS

Enviromental
triggers

In
Genaticly
susptable pt

Immune
dysregulation

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ENVIROMENTAL TRIGGERS

- TIE POD LIBA

➤ **Trauma** : Koebner phenomenon, KP” or “Isomorphic response

formation of psoriasiform lesions in uninvoived skin of psoriatic patients by injury to the skin

➤ **Infection**: guttat ps with strept infection

HIV with sever ps

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- **Endocrinal** : hypocalcemia trigger GPP
 - **PSYCOGENIC STRESS** :
 - **Other** : alcohol , smoking , obesity
 - **DRUGS** :lithium ,inf ,B blocker ,antimalarial

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GENETIC PREDISPOSITION:

- +ve family history (35%-90%)
- The risk of a child developing psoriasis...
 - if both parents had psoriasis = 41%
 - if only one parent had psoriasis = 14%
 - if just one sibling (i.e. brother or a sister) had psoriasis = 6%
- higher concordance rates of psoriasis among monozygotic compared to dizygotic twins (2-3 fold increased risk).
- Associated with certain HLA (HLA CW6) (HLA B 27)

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- **Associated with certain HLA**
 - (HLA CW6) 90% in early onset , 50% in late onset
 - (HLA B 27) pustular ps , psoriatic arthritis

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IMMUNE DYSREGULATION

- psoriasis is T-cell mediated inflammatory disease characterized by epidermal hyperplasia (keratinocyte proliferation) and angiogenesis.
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- Psoriasis is mediated helper T-cells (mainly Th1, Th17 and Th22).

Th cell	Cytokine	Actions of cytokines
Th9	TNF-α (the most important cytokine)	● Keratinocytes: increased proliferation (clinically: plaque) ● Endothelial cells: increased expression of adhesion molecules (helps extravasation of T-cells) and vascular endothelial growth factor (VEGF) → angiogenesis (clinically: erythema and Auspitz sign).
Th17	IL-17A	● Keratinocytes: increased expression of certain chemokines that result in Th17 recruitment (+ve feedback) and neutrophil chemotaxis (histopathology: Munro's and Kogoj's <i>microabscesses</i>) – also increased expression of antimicrobial peptides such as human β-defensin-1 (hBD1) → antimicrobial and <i>pro-inflammatory</i> activity.
Th22	IL-22	● Keratinocytes: increased proliferation (clinically: plaque)



Environmental triggers/genetic susceptibility

Increased production of beta-defensins, CXC-chemokines and CCL-20

Epidermal acanthosis and abnormal keratinocyte proliferation

Keratinocyte injury and increased antigen presentation

Increased production of proinflammatory (IL-23 and IL-12, IL-17, IFN- γ , TNF, and IL-22)

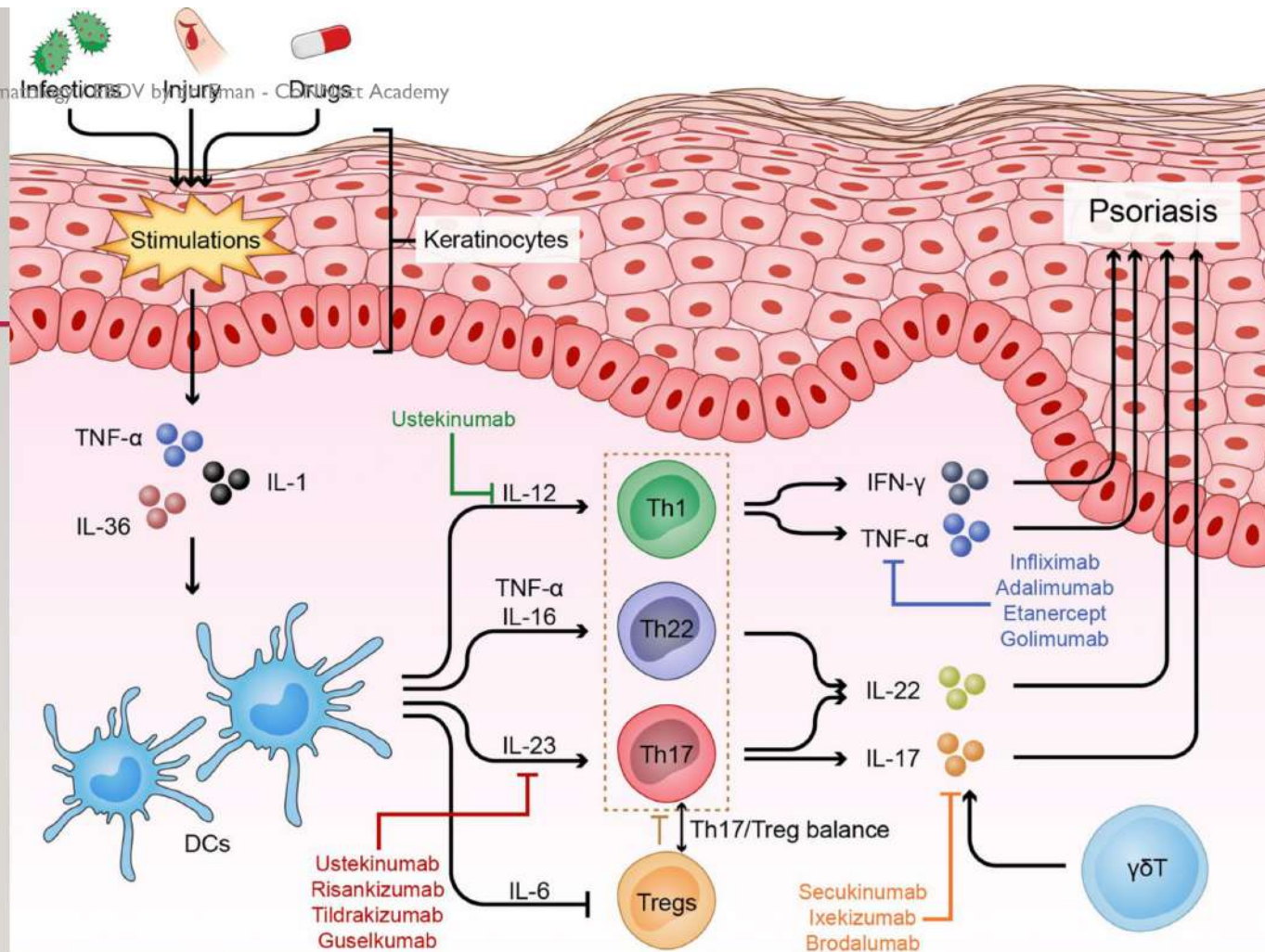
Activation of T cells by myeloid dendritic cells leading to Th1 and Th17 activation and proliferation

IL-17

IL-22

Th1 cell (TNF, IFN)

Th17 cell (IL-23)



23 WHAT IS THE RESULT OF IMMUNE DYSREGULATION

- Increased **proliferation** under the effect of keratinocyte growth factor “KGF” (turnover cell cycle is shortened to about 1 week instead of the normal 26 days).
 - Abnormal differentiation / keratinization of granular and horny cell layer (**increased expression of K6, K16, K17 instead of K1, K10**).
 - Cytokine production e.g. IL-1 (↑ selectin expression and lymphocyte recruitment) and VEGF (dilated tortuous dermal capillaries).

24 CLINICAL FEATURE

- the classic clinical finding of any type of psoriasis is erythematous scaly papules and plaques
- 1. **Erythema** (due to vascular angiogenesis resulting in elongated dilated dermal papillary capillaries).
 2. **Epidermal thickening** (Papules and plaques) (due to epidermal acanthosis “keratinocyte proliferation” and cellular infiltrates).
 3. **Scales** (due to abnormal keratinization and rapid cell turnover



25 HISTOPATHOLOGY

- epidermis

1. Parakeratosis
2. Absence granular cell layer
3. Regular acanthosis (test tube appearance) with clubbing
4. Supra papillar thinning
5. Exocytosis of some lymphocytes
6. ,neutrophils(munro ,kogoj microabsceses)

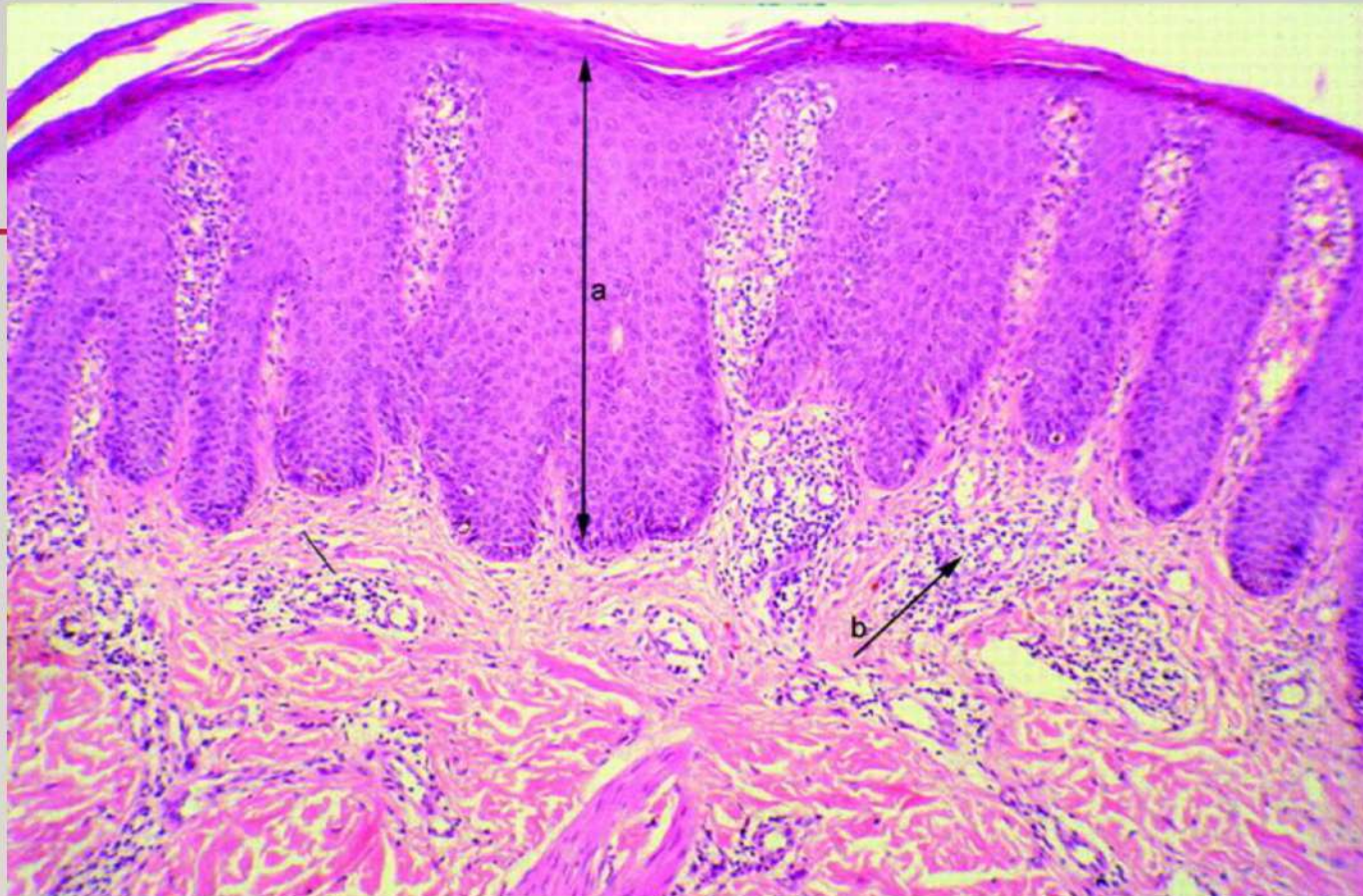


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- **Dermis**

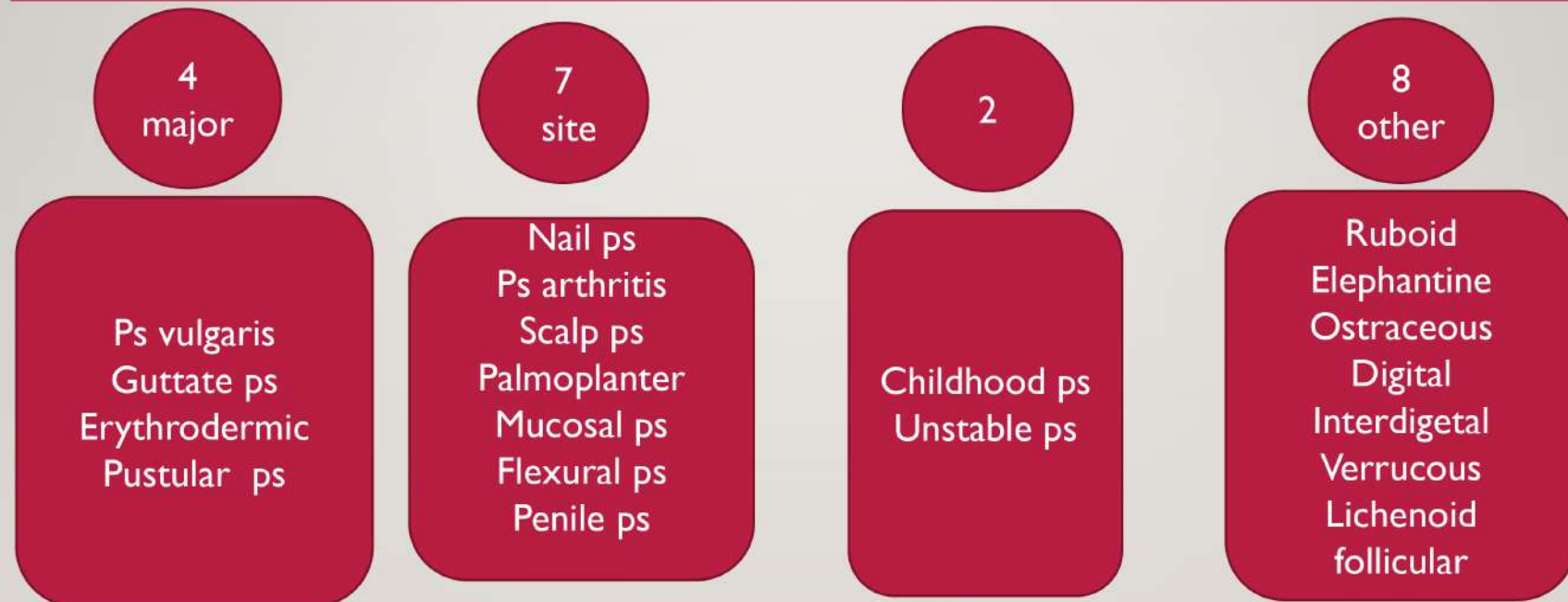
- o Increased number and length of dilated tortuous capillaries,
- o Edema at the tips of the dermal papillae (club-shaped),
- o Mixed perivascular infiltrate of lymphocytes, macrophages and neutrophils.

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VARIANTS OF PSORIASIS



29 I - PSORIASIS VULGARIS



- Most common
- Well defined erythematous plaque –silvery white scale.
- Elbowe knee , perisacrum
- Dd :hypertrophic lp, DM, mf , eczema

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2- GUTTATE PS



- Children
- URT strept infection
- Multiple small papules with thin scales
- Widely disseminated on trunk
- DD : PR, PLC ,2RY SYPHILIS

31 3-ERYTHRODERMIC PS

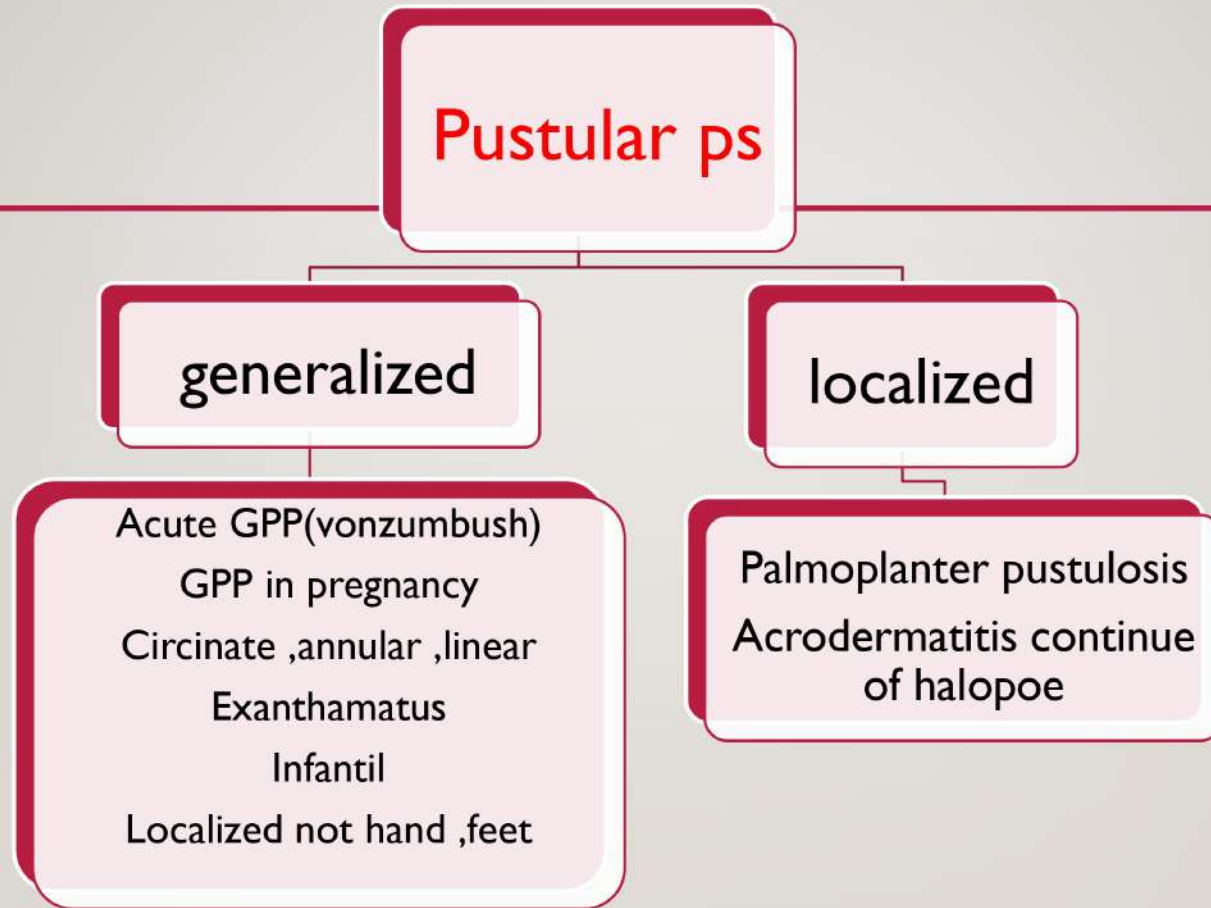


Figure 6: Erythrodermic psoriasis in an 11-year-old boy



- GENERALIZED ERYTHEMA , SCALING >90%
- DD: erythroderma

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33 THANKYOU

